

# AWS IAM ko EKS me Link Karne ke liye OIDC ka Use (IRSA Explained)

## Overview

EKS me OIDC ka use mainly **IAM Roles for Service Accounts (IRSA)** setup karne ke liye hota hai.

Isse Kubernetes pod directly AWS IAM role assume kar sakta hai — bina EC2 node ke IAM role use kiye.

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## Step-by-Step Flow

### 1. EKS Cluster OIDC Provider Create karta hai

Jab EKS cluster banta hai, uska ek OIDC issuer URL hota hai.

Example:

```
https://oidc.eks.ap-south-1.amazonaws.com/id/EXAMPLE123
```

Ye Kubernetes service account tokens ko verify karta hai.

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### 2. AWS IAM me OIDC Provider Associate karte hai

Command:

```
eksctl utils associate-iam-oidc-provider  
  --region ap-south-1  
  --cluster my-cluster  
  --approve
```

Isse AWS trust karta hai ki:

“Ye token EKS cluster se hi aaya hai.”

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### 3. IAM Role Create karte hai with Trust Policy

Example Trust Relationship:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Federated": "arn:aws:iam::123456789012:oidc-provider/oidc.eks.ap-
south-1.amazonaws.com/id/EXAMPLE123"
      },
      "Action": "sts:AssumeRoleWithWebIdentity",
      "Condition": {
        "StringEquals": {
          "oidc.eks.ap-south-1.amazonaws.com/id/EXAMPLE123:sub":
            "system:serviceaccount:default:s3-sa"
        }
      }
    }
  ]
}
```

#### Meaning

- Federated → OIDC provider
- sub → Kubernetes service account

Yaha sirf:

```
system:serviceaccount:default:s3-sa
```

wala service account IAM role assume kar sakta hai.

### 4. Kubernetes Service Account ko IAM Role se Map karte hai

YAML Example:

```
apiVersion: v1
kind: ServiceAccount
metadata:
```

```
name: s3-sa
namespace: default
annotations:
  eks.amazonaws.com/role-arn: arn:aws:iam::123456789012:role/S3ReadRole
```

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## 5. Pod Service Account Use karega

```
spec:
  serviceAccountName: s3-sa
```

Ab pod ke andar:

```
aws s3 ls
```

run karega according to IAM permissions.

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## Real Benefit

### Without OIDC / IRSA

- Sab pods node IAM role use karte
- Security risk hota
- Excess permissions mil sakti hai

### With OIDC / IRSA

- Har pod ka separate IAM role
  - Least privilege access
  - More secure
  - Better auditing
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## Architecture Flow

```
Pod
↓
Service Account
↓
```

```
OIDC Token
↓
AWS STS AssumeRoleWithWebIdentity
↓
IAM Role
↓
Access AWS Services (S3, DynamoDB, Secrets Manager etc.)
```

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## Common AWS Services Used with IRSA

- Amazon S3
  - DynamoDB
  - Secrets Manager
  - SQS
  - SNS
  - CloudWatch
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## Important Interview Points

### What is OIDC in EKS?

OIDC is used to establish trust between Kubernetes service accounts and AWS IAM.

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### Why use IRSA?

To provide pod-level IAM permissions instead of node-level permissions.

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### Which AWS API is used?

```
sts:AssumeRoleWithWebIdentity
```

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### Main Security Benefit

Least privilege access for pods.

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## One-Line Interview Answer

"OIDC in EKS is used to establish trust between Kubernetes service accounts and AWS IAM, enabling pods to securely assume IAM roles using IRSA without relying on node IAM roles."